

Notes: Daniel, Grinblatt, Titman, & Wermers (JF, 1997)

Measuring Mutual Fund Performance with Characteristic-Based Benchmarks

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1 Big picture

- **Question.** Can active mutual fund managers pick stocks that outperform simple passive strategies matched on stock characteristics?
- **Core problem.** Conventional fund performance measures can mistake style exposures for skill. A fund that buys small, high book-to-market, or momentum stocks can look skilled if the benchmark does not match those characteristics.
- **Contribution.** The paper builds characteristic-based benchmarks from 125 passive portfolios sorted on size, book-to-market, and prior-year return. It then compares each stock held by a fund to a passive portfolio of stocks with the same characteristics.
- **Data innovation.** Instead of evaluating only realized fund returns, the paper uses quarterly mutual fund holdings from CDA Investment Technologies for more than 2,500 equity funds from 1975 to 1994.
- **Three-part decomposition.** Fund returns are decomposed into Average Style (AS), Characteristic Selectivity (CS), and Characteristic Timing (CT).
- **Selectivity result.** The average fund has positive CS of about 0.77 percent per year over 1975-1994. This means fund holdings outperform stocks with the same size, book-to-market, and momentum characteristics by roughly 77 bps per year before costs.
- **Timing result.** Funds show essentially no ability to time characteristics. CT is statistically insignificant for all major investment-objective groups over the full sample.
- **Style result.** Mutual funds hold relatively large stocks, median book-to-market stocks, and above-median momentum stocks. Aggressive-growth funds hold smaller, lower book-to-market, higher-momentum stocks.
- **Why GT looks stronger.** The Grinblatt-Titman measure shows a larger performance number, but much of it is linked to momentum exposure that the characteristic benchmark controls for.
- **Persistence result.** Hypothetical gross returns show persistence, but benchmark-adjusted characteristic returns do not. This supports the view that performance persistence mainly reflects simple momentum/style effects rather than persistent stock-picking skill.
- **Economic takeaway.** Active managers appear to have modest stock-selection ability before expenses, especially aggressive-growth and growth funds, but the gains are small and roughly comparable to active management fees.

2 Data and measurement

2.1 Why holdings data matter

- Most fund performance studies evaluate actual investor returns. This paper instead evaluates the stocks that funds actually hold.
- Holdings data make it possible to construct benchmarks that match the investment style of the portfolio at the stock level.
- Hypothetical returns from holdings exclude fees, expenses, and trading costs. This is useful for measuring manager skill before implementation costs.
- The paper can separate whether a manager earns returns from holding a style, timing a style, or selecting better stocks within a style.

Interpretation. The holdings-based approach asks whether managers choose good securities, not whether investors earn positive net returns after all costs.

2.2 Mutual fund holdings sample

- The fund holdings data are from CDA Investment Technologies.
- The sample includes quarterly equity holdings for virtually all equity mutual funds that existed between December 31, 1974 and December 31, 1994.
- The raw database contains more than 5,000 funds. The paper excludes sector funds, bond funds, preferred funds, and funds without a general equity objective.
- The final holdings universe contains over 2,500 equity mutual funds.
- The database is designed not to require survival, which reduces survivorship bias relative to samples of live funds only.

Interpretation. The paper improves on earlier holdings-based studies by using a much larger, longer, and more complete sample.

2.3 Investment objectives

- The fund categories are based on self-declared investment objectives.
- The main categories are aggressive growth, growth, growth-income, and balanced/income.
- Before June 30, 1980, the paper uses the December 31, 1974 objective to classify funds.
- Balanced and income funds are combined because each group is relatively small.

Interpretation. The objective categories let the authors ask whether performance differs across styles, especially aggressive-growth versus more conservative funds.

2.4 Stock characteristics and benchmark portfolios

- Each stock is classified each year by three characteristics: market capitalization, book-to-market equity, and prior-year return.
- For each characteristic, stocks are placed into quintiles.
- The three-way sort creates $5 \times 5 \times 5 = 125$ passive benchmark portfolios.
- Quintile breakpoints are based on NYSE stocks, while NYSE, AMEX, and Nasdaq stocks are included in the portfolios.

- The 125 passive portfolios are value-weighted to avoid putting too much weight on very small stocks and to reduce rebalancing biases.

Interpretation. The benchmark for a stock is not the market index. It is a portfolio of stocks with similar size, value/growth, and momentum characteristics.

2.5 Hypothetical fund returns

- At the start of each quarter, the paper forms the portfolio implied by the fund's reported holdings.
- That portfolio is held until the start of the next quarter, when weights are updated using the next holdings report.
- If a quarterly report is missing, the most recent available holdings are used, adjusted for events such as stock splits.
- These returns are called hypothetical or gross holdings-based returns.

Interpretation. The hypothetical returns approximate what the fund manager's stock choices earned before fund expenses and trading costs.

2.6 Main performance objects

- **Gross return:** the buy-and-hold return implied by reported fund holdings.
- **GT performance:** a holdings-based measure from Grinblatt and Titman (1993), using prior holdings as a benchmark.
- **CS performance:** characteristic selectivity, comparing each stock to its matched characteristic portfolio.
- **CT performance:** characteristic timing, measuring whether the fund shifts style weights before those styles do well.
- **AS return:** average style return, measuring the return from systematically holding particular characteristics.
- **Jensen measures:** one-factor CAPM alpha and Carhart four-factor alpha.

Interpretation. CS is the most direct measure of stock-selection skill in the paper. CT measures style timing, and AS measures style exposure.

3 Models and empirical specifications

3.1 The benchmark logic

The overall idea is:

fund holdings return – matched characteristic return = manager skill within characteristics.

Interpretation. A manager should not get credit merely for buying small-cap, value, or momentum stocks if those characteristics perform well mechanically.

3.2 Characteristic Selectivity (CS)

For fund holdings at month $t - 1$, the monthly CS component is

$$CS_t = \sum_{j=1}^N w_{j,t-1} \left(R_{j,t} - R_t^{b(j,t-1)} \right), \quad (1)$$

where $w_{j,t-1}$ is the fund weight on stock j , $R_{j,t}$ is the stock return, and $R_t^{b(j,t-1)}$ is the return of the passive benchmark portfolio matched to stock j at time $t - 1$.

Interpretation. CS is positive when the stocks selected by the manager outperform stocks with the same size, book-to-market, and momentum characteristics.

3.3 Characteristic Timing (CT)

The monthly CT component compares the current characteristic exposure with the lagged characteristic exposure:

$$CT_t = \sum_{j=1}^N \left[w_{j,t-1} R_t^{b(j,t-1)} - w_{j,t-13} R_t^{b(j,t-13)} \right]. \quad (2)$$

Interpretation. CT is positive if the manager increases exposure to a characteristic shortly before that characteristic earns high returns. The one-year lag removes returns due merely to long-run style exposure.

3.4 Average Style (AS)

The monthly AS component is

$$AS_t = \sum_{j=1}^N w_{j,t-13} R_t^{b(j,t-13)}. \quad (3)$$

Interpretation. AS captures returns from the fund's persistent style. For example, a fund that consistently holds momentum stocks earns high AS when momentum strategies perform well.

3.5 Approximate decomposition

Ignoring minor implementation differences due to data requirements, total holdings-based fund return can be decomposed as

$$R_t^{fund} \approx AS_t + CT_t + CS_t. \quad (4)$$

Interpretation. This is the paper's main conceptual advance: fund returns can be separated into style exposure, style timing, and security selection.

3.6 GT measure

The Grinblatt-Titman monthly measure is

$$GT_t = \sum_{j=1}^N (w_{j,t-1} - w_{j,t-13}) R_{j,t}. \quad (5)$$

Interpretation. GT measures returns from changes in portfolio weights over the previous year. It can be influenced by momentum because funds that recently increased weights in past winners may appear to outperform even if the benchmark does not adjust for prior-year returns.

3.7 Carhart four-factor Jensen measure

The Carhart regression is

$$R_{j,t} - RF_t = \alpha_j + b_j RMRF_t + s_j SMB_t + h_j HML_t + p_j PR1YR_t + \epsilon_{j,t}. \quad (6)$$

- $RMRF$ is the market excess return.
- SMB is small minus big.
- HML is high book-to-market minus low book-to-market.
- $PR1YR$ is high prior-year return minus low prior-year return.

Interpretation. This is a factor-regression benchmark. It adjusts for covariance with factor portfolios rather than matching each stock directly on characteristics.

3.8 CAPM one-factor Jensen measure

The one-factor Jensen alpha uses

$$R_{j,t} - RF_t = \alpha_j + \beta_j (RM_t - RF_t) + \epsilon_{j,t}. \quad (7)$$

Interpretation. This is the traditional market beta adjustment. The paper shows that it is less informative than characteristic matching for this setting.

4 Identification and empirical design

4.1 What identifies selectivity

- CS compares each stock held by a fund with stocks that have similar size, book-to-market, and prior-year return.
- The identifying assumption is that these three characteristics capture the relevant passive style exposures that otherwise explain cross-sectional returns.
- Any remaining average return difference is interpreted as selectivity.

Interpretation. The design is strongest if the major return anomalies are captured by the three characteristics and if the matched portfolios are good substitutes for the stocks held by funds.

4.2 Why holdings data improve the test

- Holdings reveal what stocks the manager selected.
- Fund returns alone can be contaminated by fees, expenses, trading costs, flows, and unobserved changes in positions.
- Holdings allow benchmark matching at the stock level instead of estimating a fund-level factor exposure from returns.
- Characteristic matching usually explains more return variance and lowers standard errors relative to factor models when holdings are available.

Interpretation. The paper's goal is not to estimate investor net returns; it is to test whether managers choose stocks that outperform matched stocks.

4.3 Addressing survivorship bias

- The CDA holdings sample is designed to include funds that existed during each quarter, not only funds that survived to the end of the sample.
- This matters because poor-performing funds are more likely to close or merge.
- The performance tests are therefore less contaminated by survivorship than samples of live funds only.

Interpretation. A no-survival-requirement sample is important for measuring average skill rather than skill conditional on survival.

4.4 Why the timing test matters

- A fund can outperform by holding a profitable style persistently or by timing when that style will be profitable.
- CT isolates the timing component by comparing current characteristic exposures with exposures one year earlier.
- The paper finds no evidence that funds can time size, value, or momentum characteristics.

Interpretation. Fund managers seem better at selecting stocks within styles than predicting when styles will outperform.

4.5 Why persistence tests matter

- Previous evidence of mutual fund performance persistence could reflect true persistent skill.

- It could also reflect persistent style exposures, especially momentum.
- The paper sorts funds by past gross return and then examines whether benchmark-adjusted returns continue to outperform.
- Persistence is visible in gross returns but not in CS performance.

Interpretation. The evidence suggests that past winners held stocks with characteristics that continued to perform well, not necessarily that they had persistent stock-picking ability.

4.6 What the paper cannot fully prove

- Characteristic benchmarks are only as good as the characteristics they match on.
- The analysis is pre-cost for the holdings-based measures; investors care about net returns after expenses and trading costs.
- Fund holdings are observed quarterly, not continuously.
- The characteristic approach cannot be implemented without holdings data.
- Positive CS does not identify the exact source of skill: private information, research ability, transaction skill, or other selection advantages.

Interpretation. The paper provides strong evidence of modest pre-cost selectivity, but not necessarily of economically large net alpha for investors.

5 Tables and figures

The paper has no graphs. The original tables are placed close to their discussion and, where possible, related tables are grouped to save space.

5.1 Table I: Characteristic style analysis

Read:

- All funds have average size quintile 3.97, meaning mutual funds hold larger-than-average stocks.
- The average book-to-market quintile is 2.87, close to the median.
- The average momentum quintile is 3.11, indicating above-median prior-year returns.
- Aggressive-growth funds hold smaller stocks (size 3.20), lower book-to-market stocks (2.50), and higher-momentum stocks (3.31).
- Growth-income and balanced/income funds hold larger and higher book-to-market stocks.

Economic message: The table shows that stated fund objectives map into stock characteristics, but the differences across categories are not enormous. This motivates stock-level benchmark matching rather than relying only on fund objective labels.

Table I
Characteristic (Style) Analysis

In the table below, the average characteristic (or style) is shown for funds in different investment-objective groups. To calculate this average characteristic, the quintile benchmark portfolio number (1 through 5) is identified, each year, for each stock held by each mutual fund for each of three different characteristics. These characteristics are market capitalization, book-to-market equity value, and prior-year return. Size portfolio 1 consists of small stocks, book-to-market portfolio 1 consists of low book-to-market stocks, and momentum portfolio 1 consists of low prior-year return stocks. Then, for each mutual fund, the portfolio-weighted average benchmark portfolio number is computed each year (for each characteristic). Finally, the fund average benchmark portfolio numbers are averaged across all funds existing in each category each year, and the resulting time-series is averaged over all years from 1975 to 1994. Before June 30, 1980, the investment objective of December 31, 1974 was used to classify funds.

	Average Size Quintile	Average Book-to- Market Quintile	Average Momentum Quintile
All funds ^a	3.97	2.87	3.11
Aggressive growth	3.20	2.50	3.31
Growth	3.99	2.75	3.10
Growth and income	4.34	3.13	3.03
Balanced or income	4.34	3.29	3.03

^a Limited to funds having an investment objective of aggressive-growth, growth, growth-income, income, or balanced.

Table 1: Table I: directly cropped from the original paper.

5.2 Table II: Mutual fund returns, 1975-1994

Read:

- The average gross holdings-based fund return is 17.03 percent per year over 1975-1994.
- The GT measure averages 1.91 percent per year and is highly significant.
- The CS measure averages 0.77 percent per year and is significant at the 5 percent level.
- In 1975-1984, CS is 1.15 percent per year; in 1985-1994, CS falls to 0.39 percent and is not significant.
- The stronger early-period performance suggests either less efficient markets or a changing composition of active managers over time.

Economic message: Mutual funds appear to have modest pre-cost stock-selection ability, but much weaker than what GT alone would suggest.

Table II
Mutual Fund Returns, 1975–1994

For each year (row) below, the average buy-and-hold monthly return (annualized to percent per year) is presented for the Center for Research in Security Prices value weighted New York Stock Exchange/American Stock Exchange (CRSP VW) portfolio, CRSP equally weighted NYSE/AMEX portfolio (CRSP EW), and the equally weighted portfolio of all funds existing during the year with a self-declared investment objective of aggressive-growth, growth, growth-income, income, or balanced. In addition, the total number of funds in these five investment-objective categories at the end of each year is shown. For the period before 1980, all funds (regardless of investment objective) are included in the averages. Finally, the GT and CS (Characteristic Selectivity) performance measures are presented for the equally weighted portfolio of the same funds. The GT measure is calculated by subtracting the time t return of the portfolio held at month $t - 13$ from the time t return of the portfolio held at $t - 1$. The CS measure is the difference between the time t return of the portfolio held at time $t - 1$ and the time t return of the time $t - 1$ matching control portfolio. All portfolios are rebalanced monthly. Time-series t -statistics are shown in parentheses.

Year	CRSP VW w/Dividends	CRSP EW w/Dividends	Equally Weighted Portfolio of Funds*			
			Number	Gross Return	GT Performance	CS Performance
1975	34.00	57.23	383	36.28	NA	0.78 (0.49)
1976	24.64	43.53	383	26.97	1.47 (1.33)	0.38 (0.27)
1977	-3.81	14.90	588	0.82	2.01* (1.93)	0.91 (0.89)
1978	9.00	21.90	547	14.19	2.21 (1.17)	3.00* (2.02)
1979	22.33	35.84	545	29.50	2.12 (1.75)	2.50* (1.96)
1980	30.50	32.73	485	34.60	3.67 (1.65)	1.06 (0.55)
1981	-3.48	5.73	479	-1.39	0.25 (0.13)	0.72 (0.60)
1982	20.21	27.40	481	24.64	5.39*** (3.47)	2.59 (1.37)
1983	21.37	32.54	481	22.40	0.75 (0.46)	0.99 (0.93)
1984	5.86	-2.07	508	0.96	0.25 (0.29)	-1.40 (-0.78)
1985	28.11	24.85	520	32.91	1.99 (1.63)	-0.23 (-0.16)
1986	17.07	12.46	582	15.67	1.55 (0.98)	0.06 (0.04)

Table II—Continued

Year	CRSP VW w/Dividends	CRSP EW w/Dividends	Equally Weighted Portfolio of Funds*			
			Number	Gross Return	GT Performance	CS Performance
1987	7.29	0.05	653	8.01	2.20 (1.00)	2.04 (0.74)
1988	16.66	19.59	735	17.43	-0.85 (-0.47)	-0.51 (-0.30)
1989	26.61	15.53	786	26.34	4.04*** (4.23)	0.28 (0.30)
1990	-3.49	-21.35	829	-5.95	2.87 (1.75)	0.11 (0.05)
1991	27.95	35.27	984	34.84	3.02 (1.36)	1.15 (0.72)
1992	7.94	20.10	1,049	9.73	0.18 (0.17)	0.05 (0.04)
1993	10.61	21.16	1,421	13.12	3.65 (1.60)	0.75 (0.60)
1994	0.14	-2.87	1,973	-0.44	-0.40 (-0.52)	0.19 (0.17)
1975–1994 ^b	14.98	19.73		17.03	1.91*** (5.19)	0.77** (2.23)
1975–1984 ^b	16.06	26.97		18.90	2.01*** (3.88)	1.15** (2.45)
1985–1994	13.89	12.48		15.17	1.82*** (3.47)	0.39 (0.77)

* Significant at the 10 percent significance level.

** Significant at the 5 percent significance level.

*** Significant at the 1 percent significance level.

5.3 Table III: Carhart-adjusted measures

Read:

- Panel A shows that GT performance loads heavily on momentum: the PR1YR coefficient is 0.11 for all funds and 0.18 for aggressive-growth funds.
- The intercept for GT falls to 0.56 percent per year after Carhart adjustment, much lower than the unadjusted GT estimate of 1.91 percent.
- Panel B shows that CS remains positive after Carhart adjustment; the all-fund intercept is 0.56 percent per year.
- Aggressive-growth funds have a Carhart-adjusted CS intercept of 1.41 percent per year.
- HML coefficients in the CS regressions are negative, consistent with characteristic matching already removing much of the value exposure.

Economic message: Much of the GT result is momentum exposure. The CS result is more robust because it already controls directly for stock characteristics.

Table III
Carhart-Adjusted Measures

The dependent variables in these regressions are the time-series of monthly GT performance components that would accrue to buying an equally weighted portfolio, each month, of all funds having the appropriate investment objective during that month. Before June 30, 1980, the investment objective of December 31, 1974 was used to classify funds. The four independent variables are the time-series of monthly returns associated (1) with a value-weighted market proxy portfolio minus T-bills (RMRF), (2) with the difference in returns between small and large capitalization stocks (SMB), (3) with the difference in returns between high and low book-to-market stocks (HML), and (4) with the difference in returns between stocks having high and low prior-year returns (PRIYR). See Carhart (1997) for further description of these variables. The intercept has been annualized to percent per year. The *t*-statistics are shown in parentheses below coefficient estimates.

Dependent Variable	Independent Variables				
	Intercept	RMRF	SMB	HML	PRIYR
Panel A. Regressions of GT Measures on Carhart Factors (1976–1994)					
All funds ^a	0.56* (1.91)	0.01 (1.40)	0.03*** (2.71)	0.01 (1.04)	0.11*** (13.11)
Aggressive-growth funds	1.76*** (3.24)	0.03*** (2.89)	0.03* (1.72)	0.01 (0.64)	0.18*** (11.70)
Growth funds	0.47 (1.43)	0.01 (1.12)	0.04*** (3.72)	0.01 (0.76)	0.12*** (13.18)
Panel B. Regressions of CS Measures on Carhart Factors (1975–1994)					
All funds ^a	0.56* (1.87)	0.04*** (6.24)	0.03*** (2.91)	−0.06*** (−5.66)	0.01 (0.93)
Aggressive-growth funds	1.41** (2.07)	0.10*** (7.26)	0.08*** (3.55)	−0.20*** (−8.65)	0.005 (0.27)
Growth funds	0.54 (1.60)	0.05*** (6.97)	0.04*** (3.92)	−0.06*** (−5.33)	0.01 (1.19)

* Statistically significant at the 10-percent level.

** Statistically significant at the 5-percent level.

*** Statistically significant at the 1-percent level.

Table 2: Table III: directly cropped from the original paper.

5.4 Table IV: Comparison of performance measures

Read:

- For all funds, GT performance is 1.94 percent per year, while CS performance is 0.79 percent per year.
- One-factor Jensen alpha is 0.60 percent per year and Carhart four-factor alpha is 0.39 percent per year, both insignificant for all funds.
- Aggressive-growth funds have the highest GT performance: 4.02 percent per year over 1975-1994.
- Aggressive-growth funds also have positive CS of 1.49 percent per year, significant at the 10 percent level.
- Growth funds have CS of 1.03 percent per year, significant at the 5 percent level.
- Growth-income and balanced/income funds have weak or near-zero CS.

Economic message: Stock-selection ability is concentrated in aggressive-growth and growth funds. More conservative categories show little selectivity once characteristics are controlled.

Table IV
Comparison of Performance Measures

In each panel below, five measures of performance are presented. To calculate the gross return, the GT performance, and the CS performance, an equally weighted portfolio of all funds existing during each month and having a self-declared investment objective as specified in each panel during the most-recent quarter-end is formed. Before June 30, 1980, the investment objective of December 31, 1974 is used. The buy-and-hold return (or performance) of that portfolio during that month is then computed, and, finally, the time-series average across all months in a given time period is computed. For the one-factor Jensen measures, the monthly buy-and-hold excess return on the equally weighted portfolio of funds was regressed on the time-series of monthly returns associated with a value-weighted market proxy portfolio minus T-bills. For the four-factor Jensen measure, the same time-series was regressed on three additional monthly time-series, associated with (1) the difference in returns between small- and large-capitalization stocks, (2) the difference in returns between high and low book-to-market stocks, and (3) the difference in returns between stocks having high and low prior-year returns. See Fama and French (1993) and Carhart (1997) for further description of these variables. All returns have been annualized to percent per year. Also, the number of funds existing at the end of each time-period is shown. Time-series *t*-statistics are shown in parentheses.

	Number	Gross Return	GT Performance	CS Performance	One-Factor Jensen Measure	Four-Factor Jensen Measure
Panel A. All Funds (Returns are Monthly, Annualized to Percent Per Year) ^a						
1975–1979 ^b	214	21.33	2.06*** (2.89)	1.58** (2.21)	2.78** (2.40)	1.44 (1.50)
1980–1984	508	16.31	2.10** (2.41)	0.79 (1.10)	0.62 (0.55)	0.98 (0.97)
1985–1989	786	20.07	1.79** (2.46)	0.33 (0.43)	−0.80 (−0.69)	0.83 (0.96)
1990–1994	1,973	10.26	1.86** (2.43)	0.45 (0.66)	−0.16 (−0.16)	−0.36 (−0.68)
1975–1994 ^c	16.99	1.94*** (5.02)	0.79** (2.20)	0.60 (1.07)	0.39 (0.87)	
Panel B. Aggressive-Growth Funds (Returns are Monthly, Annualized to Percent Per Year)						
1975–1979 ^b	59	25.43	3.44*** (2.85)	2.71* (1.99)	5.39** (2.25)	1.67 (1.01)
1980–1984	97	15.50	4.16*** (2.92)	1.07 (0.63)	−1.37 (−0.46)	1.04 (0.56)
1985–1989	150	18.09	5.20*** (3.59)	0.12 (0.06)	−4.95 (−1.58)	0.96 (0.57)
1990–1994	211	12.49	3.15** (2.63)	2.07 (1.05)	0.62 (0.21)	0.66 (0.48)
1975–1994 ^c	17.88	4.02*** (6.01)	1.49* (1.66)	−0.21 (−0.14)	0.40 (0.43)	
Panel C. Growth Funds (Returns are Monthly, Annualized to Percent Per Year)						
1975–1979 ^b	69	20.30	1.99*** (2.74)	2.21** (2.34)	1.37 (0.96)	1.20 (1.02)
1980–1984	219	16.07	2.35** (2.38)	0.85 (0.97)	0.25 (0.19)	1.20 (1.08)
1985–1989	366	20.41	1.58** (2.17)	0.65 (0.75)	−0.75 (−0.57)	1.19 (1.31)
1990–1994	1,207	10.11	1.83** (2.22)	0.42 (0.61)	−0.33 (−0.35)	−0.42 (−0.80)
1975–1994 ^c	16.72	1.93*** (4.66)	1.03** (2.43)	0.12 (0.19)	0.32 (0.63)	

Table IV—Continued

	Number	Gross Return	GT Performance	CS Performance	One-Factor Jensen Measure	Four-Factor Jensen Measure
Panel D. Growth-Income Funds (Returns are Monthly, Annualized to Percent Per Year)						
1975–1979 ^b	49	18.98	1.21** (2.40)	0.62 (1.11)	1.36* (1.75)	1.30 (1.62)
1980–1984	122	16.47	1.15 (1.61)	0.72 (1.22)	1.23 (1.46)	1.28 (1.28)
1985–1989	172	20.51	0.09 (0.17)	−0.12 (−0.35)	0.94 (1.01)	0.17 (0.18)
1990–1994	354	9.26	1.32** (2.46)	−0.48 (−1.19)	−0.34 (−0.47)	−0.91 (−1.48)
1975–1994 ^c	16.30	0.93*** (3.18)	0.19 (0.76)	0.80* (1.96)	0.24 (0.56)	
Panel E. Balanced and Income Funds (Returns are Monthly, Annualized to Percent Per Year)						
1975–1979 ^b	37	19.67	1.03** (2.32)	−0.27 (−0.45)	3.15** (2.52)	1.84 (1.63)
1980–1984	70	17.57	0.53 (0.89)	0.39 (0.39)	3.02** (2.08)	0.52 (0.43)
1985–1989	98	21.10	0.46 (1.03)	0.19 (0.30)	2.37* (1.90)	0.67 (0.63)
1990–1994	201	9.39	1.03* (1.68)	−0.23 (−0.61)	−0.35 (−0.52)	−0.67 (−1.02)
1975–1994 ^c	16.93	0.74*** (2.74)	0.02 (0.05)	2.09*** (3.40)	0.82 (1.49)	

* Significant at the 10 percent significance level.

** Significant at the 5 percent significance level.

*** Significant at the 1 percent significance level.

^a Limited to funds having an investment objective of aggressive-growth, growth, growth-in.

5.5 Table V: Tests of performance persistence

Read:

- Funds ranked in the best prior-year gross-return quintile have a following-year gross return of 18.40 percent.
- Funds in the worst quintile have 13.80 percent, so the gross-return spread is 4.60 percent per year.
- GT performance is actually lower for the best quintile than the worst quintile, producing a negative P1-P5 spread of -1.08 percent.
- CS performance does not show significant persistence: the P1-P5 spread is 0.90 percent but insignificant.
- The one-factor Jensen measure produces a significant P1-P5 spread, but this is not robust to characteristic benchmarks.

Economic message: Past winners do not persistently exhibit characteristic-adjusted selectivity. Persistence in raw returns is largely explained by characteristics such as momentum.

Table VI
Performance Attribution Analysis

In each panel below, three performance attribution components are presented for funds in different investment objective categories. These three components are calculated as follows. The “average style attribute” (AS) is calculated, for month t , by matching each stock held by a mutual fund, on month $t - 13$, with the proper control portfolio at month $t - 13$. Then, the measure for a fund is computed by applying each stock weight at $t - 13$ to the matching control portfolio return at t . The “characteristic timing attribute” (CT) is computed, for each fund, by matching stocks held at time $t - 13$ and at time $t - 1$ with the proper control portfolios at time $t - 13$ and time $t - 1$, respectively. Next, the portfolio-weighted return of the time $t - 13$ matching portfolio, at time t , is subtracted from the portfolio-weighted return of the time $t - 1$ control portfolio, also at time t . The “characteristic-selectivity attribute” (CS) is calculated as described in the notes to Table II. All numbers presented below are computed by averaging, each month, the given measure over all funds having the appropriate self-reported investment objective during that month. Before June 30, 1980, the investment objective of December 31, 1974 was used to classify the funds. The time-series average and t -statistic are presented below (t -statistic in parentheses).

	AS Attribute	CT Attribute	CS Attribute ^a
Panel A. 1976–1994 (Monthly, Annualized to Percent Per Year)			
All funds ^b	14.85	0.02 (0.11)	0.79** (2.20)
Aggressive-growth	14.89	0.23 (0.90)	1.49* (1.66)
Growth	14.56	−0.06 (−0.29)	1.03** (2.43)
Growth-income	14.75	−0.02 (−0.11)	0.19 (0.76)
Balanced or income	15.23	0.04 (0.14)	0.02 (0.05)
Panel B. 1976–1984 (Monthly, Annualized to Percent Per Year)			
All funds ^b	15.01	0.39 (1.39)	1.18** (2.34)
Aggressive-growth	15.87	0.40 (0.90)	1.89* (1.75)
Growth	14.41	0.26 (0.72)	1.53** (2.37)
Growth-income	14.35	0.51 (1.59)	0.67* (1.66)
Balanced or income	15.42	0.56 (1.06)	0.06 (0.10)
Panel C. 1985–1994 (Monthly, Annualized to Percent Per Year)			
All funds ^b	14.71	−0.31 (−1.65)	0.39 (0.77)
Aggressive-growth	14.02	0.08 (0.28)	1.09 (0.76)
Growth	14.69	−0.35 (−1.58)	0.54 (0.97)
Growth-income	15.11	−0.50** (−2.00)	−0.30 (−1.14)
Balanced or income	15.06	−0.43* (−1.87)	−0.02 (−0.06)

* Significant at the 10 percent significance level.

** Significant at the 5 percent significance level.

*** Significant at the 1 percent significance level.

Table 3: Table VI: directly cropped from the original paper.

5.6 Table VI: Performance attribution analysis

Read:

- AS returns are large and similar across objectives, around 14.9-15.2 percent per year.
- CT is economically small and statistically insignificant over the full 1976-1994 period for every category.
- All-fund CS is 0.79 percent per year and significant.
- Aggressive-growth CS is 1.49 percent per year and growth fund CS is 1.03 percent per year.
- In 1976-1984, CS is stronger; in 1985-1994, CS is weaker and mostly insignificant.

Economic message: Average fund performance mostly comes from style exposure. Selectivity is modest but positive; characteristic timing is essentially absent.

Table V

Tests of Performance Persistence

This table presents the average monthly buy-and-hold gross return over 1976–1994 (annualized to percent per year) for equally weighted portfolios of mutual funds which were ranked on their gross return during the prior year. To select these funds each year, all funds existing during the entire prior year and having an investment objective of AG, G, GI, I, or B were ranked on their average monthly return of the prior year. Quintile portfolios were formed, and the gross return for the equally weighted portfolio of funds in each quintile was measured over the following year. All mutual funds existing during a given month were included in the following-year returns calculation, even if the fund did not survive the entire year. Then, the entire sort process was repeated for the following year. Finally, the time-series average return for each portfolio was calculated. The portfolio labeled “Best” is the quintile with the highest prior-year return, while “Worst” is the portfolio with the lowest prior-year return. For example, the gross return for a strategy of buying last year’s best quintile of funds, each year, was 18.40 percent per year. Also presented are four measures of performance for the gross-returns sort strategy.

	Best P1	P2	P3	P4	Worst P5	P1 Minus P5
Gross return	18.40	16.50	15.89	14.89	13.80	4.60*** (2.63)
GT performance	1.79*** (3.96)	1.34*** (3.34)	1.62*** (4.40)	1.86*** (4.70)	2.87*** (5.41)	−1.08** (−2.00)
CS performance	1.08** (2.01)	0.99** (2.50)	0.87** (2.38)	0.55 (1.33)	0.18 (0.31)	0.90 (1.19)
One-factor Jensen measure	2.70** (2.27)	1.09 (1.62)	0.70 (1.40)	−0.21 (−0.36)	−1.51 (−1.44)	4.21** (2.39)
Four-factor Jensen measure	0.16 (0.17)	0.12 (0.22)	0.94** (2.00)	0.72 (1.25)	0.77 (0.79)	−0.61 (−0.37)

* Significant at the 10 percent significance level.

** Significant at the 5 percent significance level.

*** Significant at the 1 percent significance level.

Note: Limited to funds having an investment objective of aggressive-growth, growth, growth-income, income, or balanced.

Table VI

Performance Attribution Analysis

In each panel below, three performance attribution components are presented for funds in different investment objective categories. These three components are calculated as follows. The “average style attribute” (AS) is calculated, for month t , by matching each stock held by a mutual fund, on month $t - 13$, with the proper control portfolio at month $t - 13$. Then, the measure for a fund is computed by applying each stock weight at $t - 13$ to the matching control portfolio return at t . The “characteristic timing attribute” (CT) is computed, for each fund, by matching stocks held at time $t - 13$ and at time $t - 1$ with the proper control portfolios at time $t - 13$ and time $t - 1$, respectively. Next, the portfolio-weighted return of the time $t - 13$ matching portfolio, at time t , is subtracted from the portfolio-weighted return of the time $t - 1$ control portfolio, also at time t . The “characteristic-selectivity attribute” (CS) is calculated as described in the notes to Table II. All numbers presented below are computed by averaging, each month, the given measure over all funds having the appropriate self-reported investment objective during that month. Before June 30, 1980, the investment objective of December 31, 1974 was used to classify the funds. The time-series average and t -statistic are presented below (t -statistic in parentheses).

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All funds ^b	14.71	−0.31 (−1.65)	0.39 (0.77)
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Growth-income	15.11	−0.50** (−2.00)	−0.30 (−1.14)
Balanced or income	15.06	−0.43* (−1.87)	−0.02 (−0.06)

^a Significant at the 10 percent significance level.

** Significant at the 5 percent significance level.

*** Significant at the 1 percent significance level.

6 Short summary

- The paper develops characteristic-based mutual fund performance benchmarks.
- Benchmarks are based on 125 passive portfolios sorted by size, book-to-market, and prior-year return.
- The data cover more than 2,500 equity mutual funds from 1975 to 1994 using quarterly holdings from CDA.
- The paper evaluates hypothetical holdings-based returns before fees, expenses, and trading costs.
- The CS measure captures stock selection relative to matched stocks.
- The CT measure captures attempts to time characteristics such as value, size, and momentum.
- The AS measure captures persistent average style exposure.
- Mutual funds hold relatively large stocks, roughly median book-to-market stocks, and above-median momentum stocks.
- The average fund earns CS of about 0.77 percent per year over the full sample.
- CS is stronger in the first half of the sample and weaker in the second half.
- Aggressive-growth and growth funds show the strongest selectivity.
- Growth-income and balanced/income funds show little selectivity.
- GT performance is much larger than CS because GT is influenced by momentum exposure.
- Carhart-adjusted results confirm that momentum explains much of GT performance.
- CT is insignificant, implying little evidence that funds time characteristics successfully.
- Raw fund return persistence is present, but characteristic-adjusted persistence is not.

- The evidence supports modest pre-cost skill but does not show that the average fund produces large net alpha.
- The results are consistent with an equilibrium in which informed managers earn enough before costs to cover active-management fees.

7 Conclusion

7.1 Core conclusion

Daniel, Grinblatt, Titman, and Wermers show that mutual fund managers have some stock-selection ability when evaluated using holdings-based characteristic benchmarks. The average fund beats matched characteristic portfolios by less than one percent per year before costs, and the strongest evidence comes from aggressive-growth and growth funds.

7.2 Economic intuition

A fund should not receive credit for holding stocks with characteristics that mechanically earn high returns. The key question is whether the stocks selected by the fund outperform other stocks with the same characteristics. Characteristic-based benchmarks answer this question directly by matching each holding to stocks with similar size, book-to-market, and momentum.

7.3 Takeaway

The paper changes the benchmark from a market or factor portfolio to a stock-level characteristic match. This makes the performance evaluation sharper. The resulting message is balanced: active mutual funds show modest pre-cost selectivity, little characteristic timing ability, and little true benchmark-adjusted persistence.